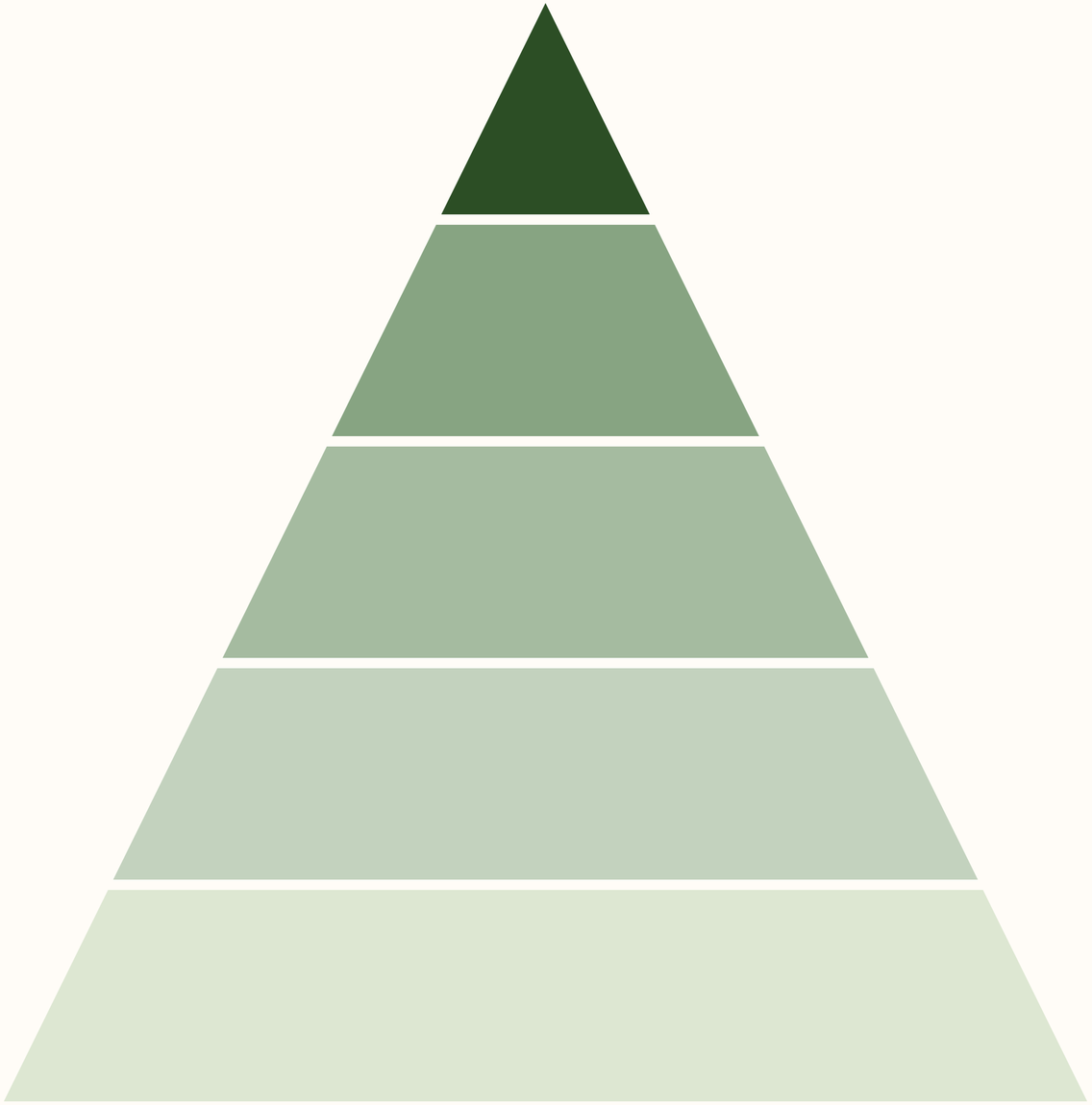
Optimal

# Iron Deficiency in *Women*



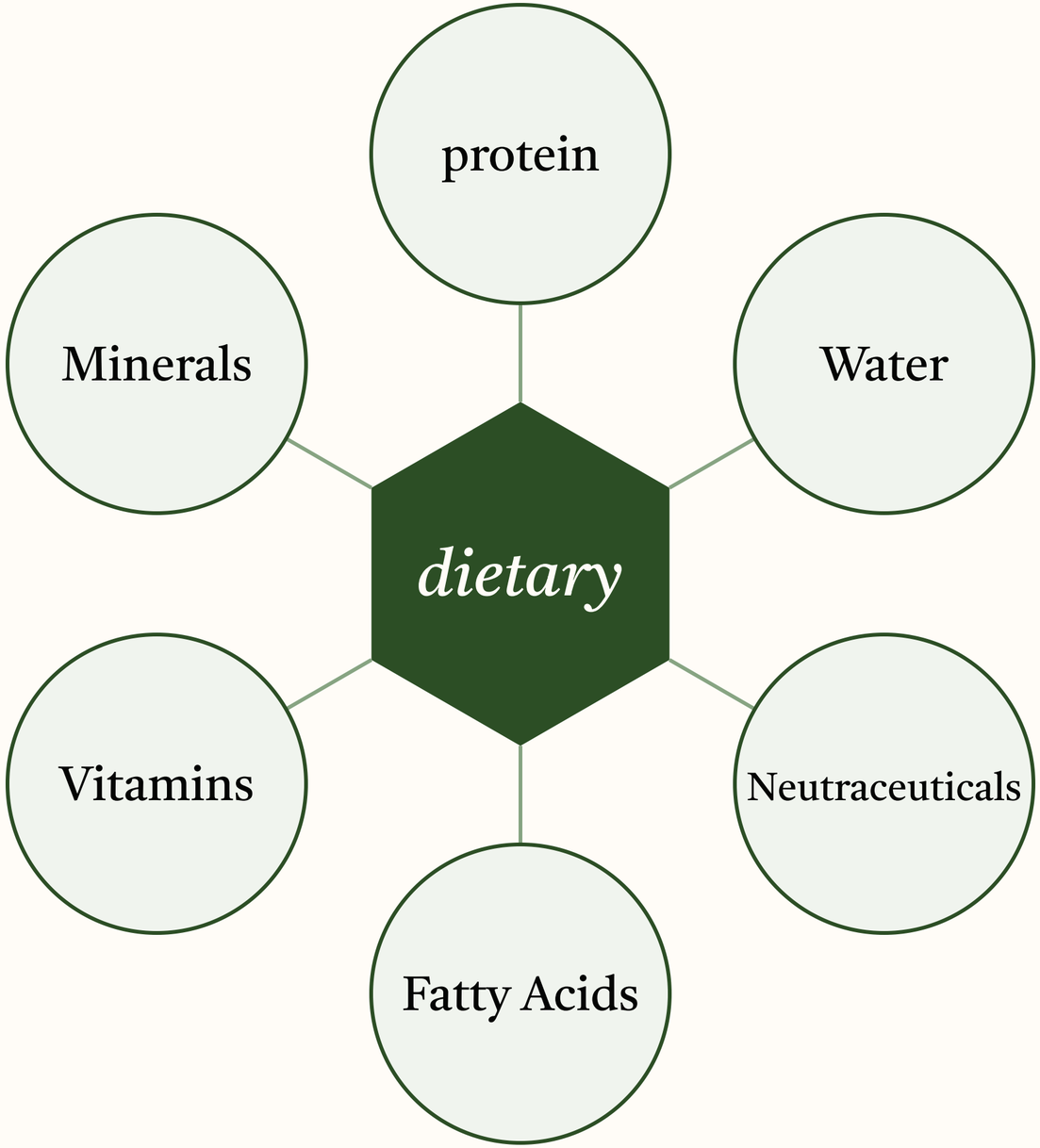
Neutraceuticals, Additional Supplements

Protein, Carb, Fat intake, goal specific activity

Quantity of Activity

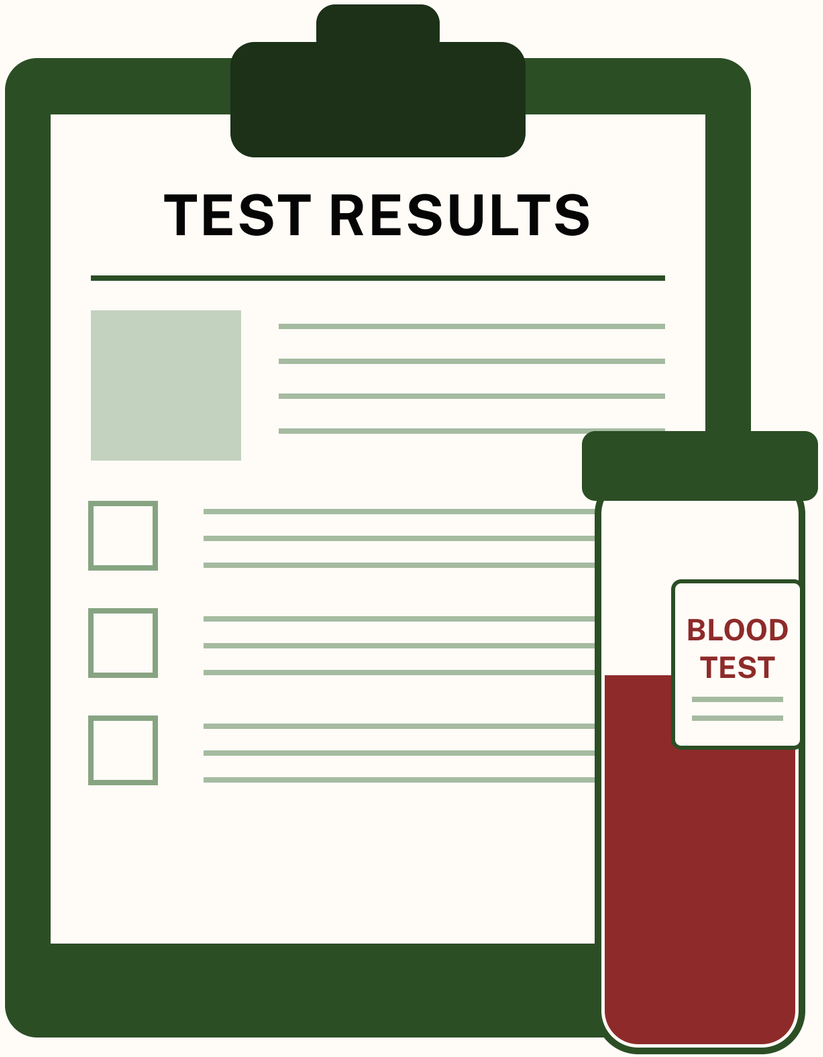
Mental Well Being

Adequate rest, caloric intake, hydration, safety


protein
Minerals
Water
dietary
Vitamins
Neutraceuticals
Fatty Acids

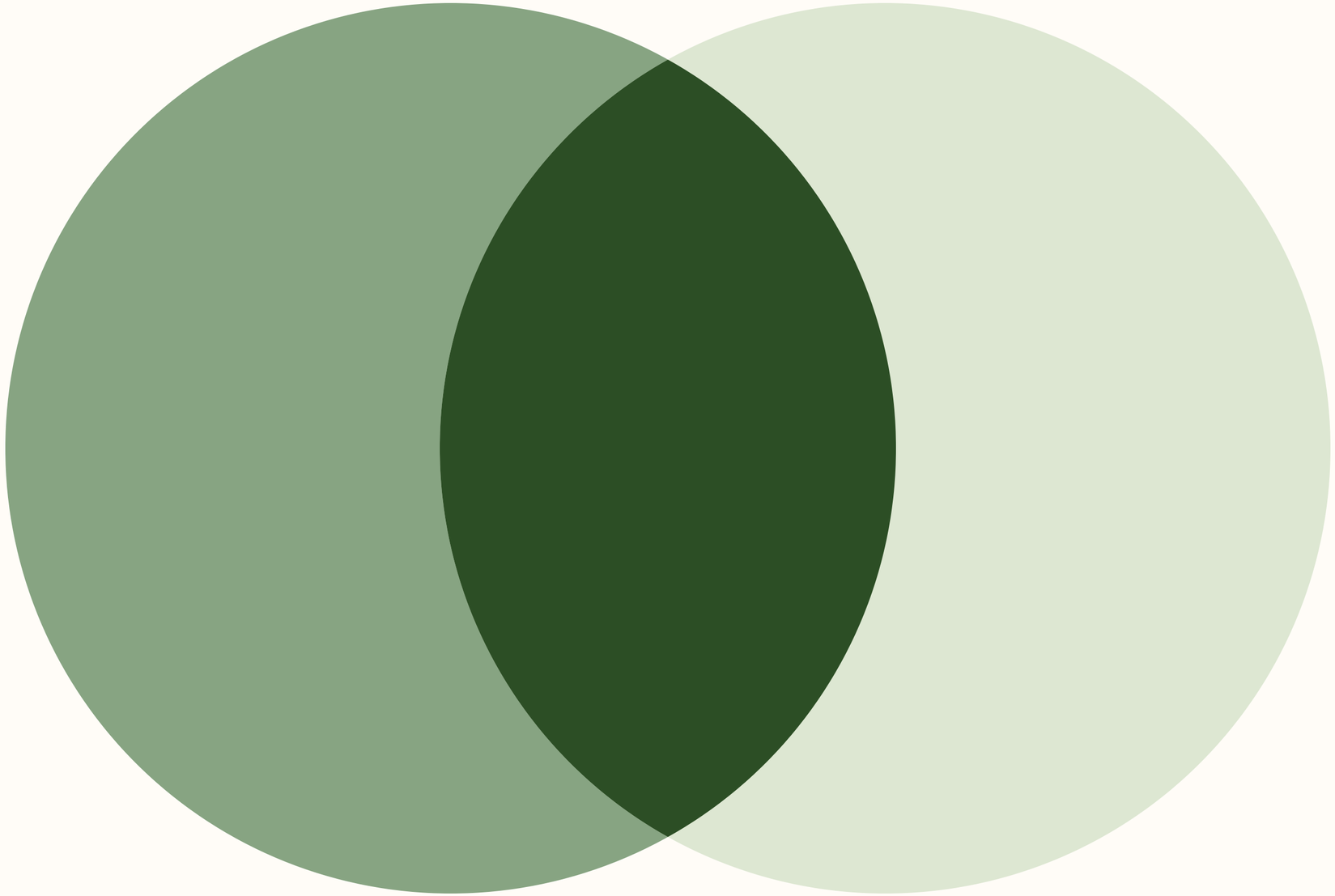
# *Agenda*

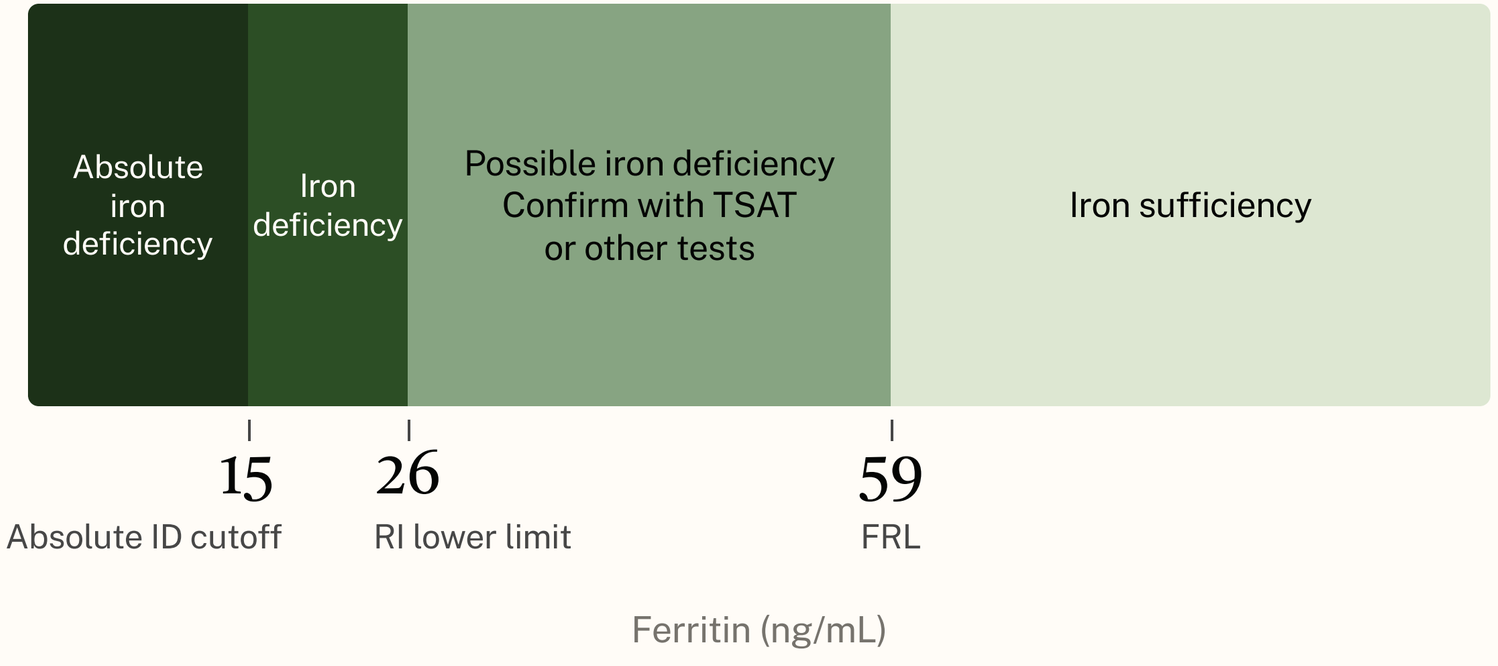
- How we evaluate iron stores
- Iron levels
- Consequences of high/low iron
- Treatment Options

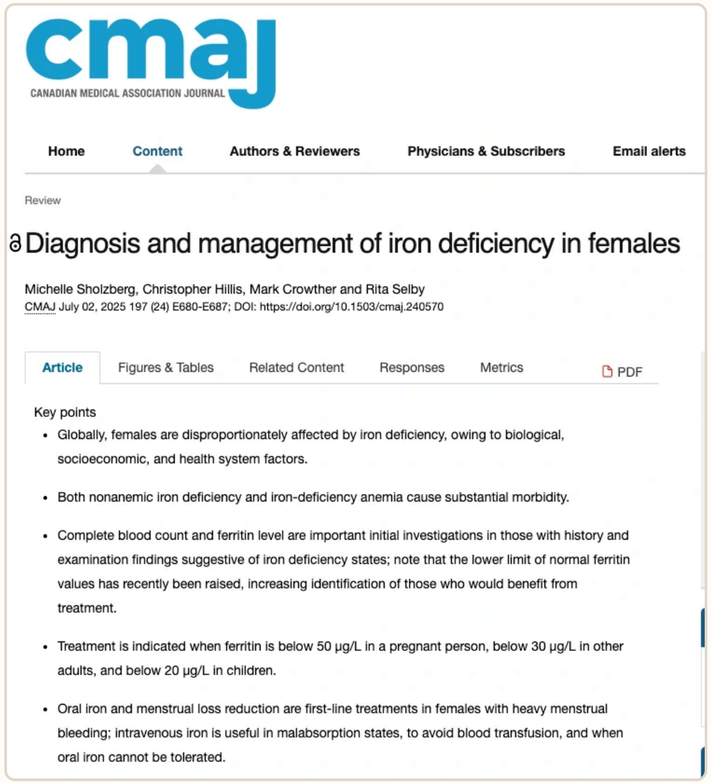

TEST RESULTS
BLOOD
TEST




TEST RE
Ferritin





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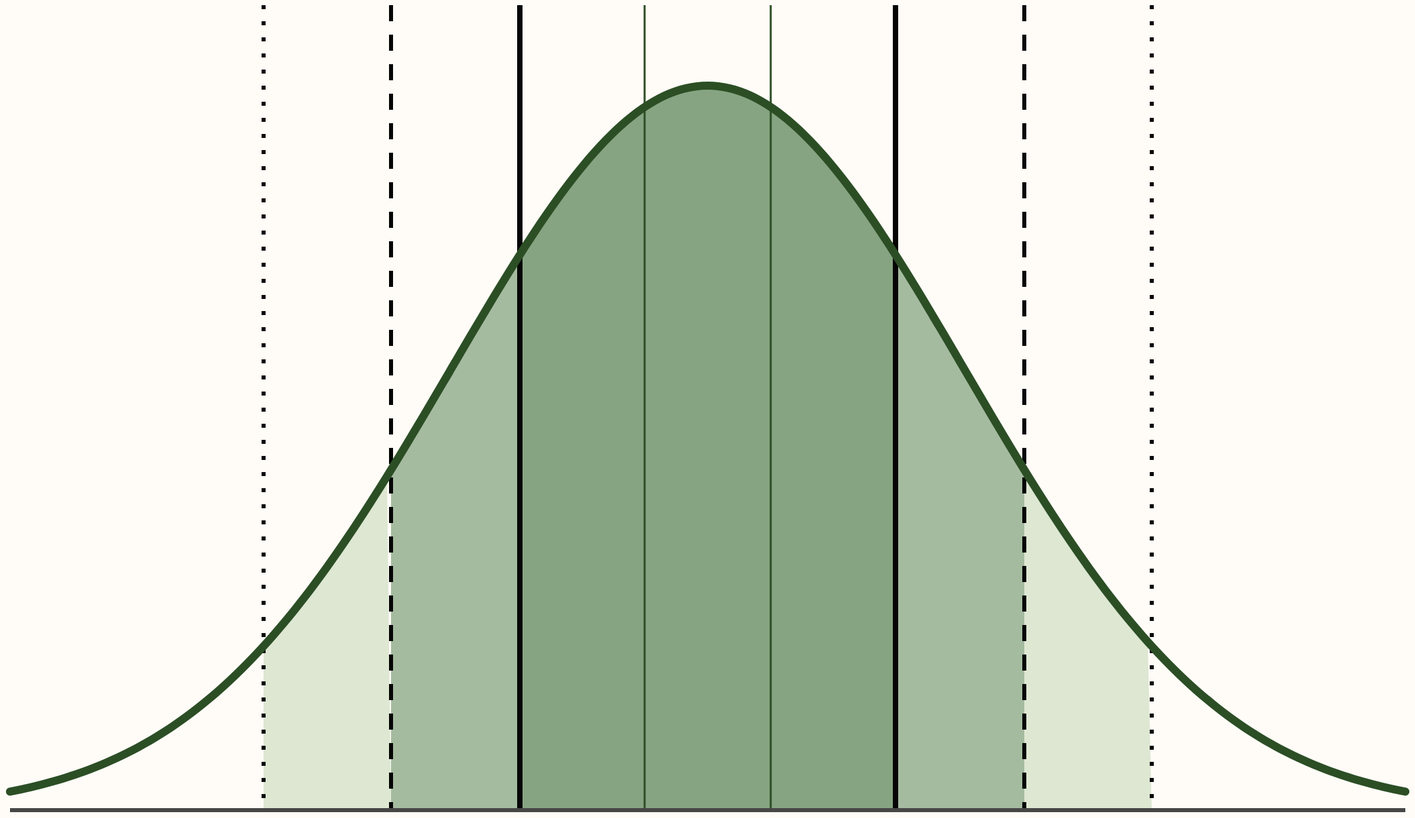
# Diagnosis and management of iron deficiency in females

Michelle Sholzberg, Christopher Hillis, Mark Crowther and Rita Selby
CMAJ July 02, 2025 197 (24) E680-E687; DOI: https://doi.org/10.1503/cmaj.240570

Article Figures & Tables Related Content Responses Metrics PDF

Key points

- Globally, females are disproportionately affected by iron deficiency, owing to biological, socioeconomic, and health system factors.
- Both nonanemic iron deficiency and iron-deficiency anemia cause substantial morbidity.
- Complete blood count and ferritin level are important initial investigations in those with history and examination findings suggestive of iron deficiency states; note that the lower limit of normal ferritin values has recently been raised, increasing identification of those who would benefit from treatment.
- Treatment is indicated when ferritin is below 50 µg/L in a pregnant person, below 30 µg/L in other adults, and below 20 µg/L in children.
- Oral iron and menstrual loss reduction are first-line treatments in females with heavy menstrual bleeding; intravenous iron is useful in malabsorption states, to avoid blood transfusion, and when oral iron cannot be tolerated.



Review > Clin Med (Lond). 2021 Mar;21(2):107-113. doi: 10.7861/clinmed.2020-0582.

# Iron deficiency without anaemia: a diagnosis that matters

Abdulrahman Al-Naseem [1], Abdelrahman Sallam [2], Shamim Choudhury [2], Jecko Thachil [3]

Affiliations + expand

PMID: 33762368 PMCID: PMC8002799 DOI: 10.7861/clinmed.2020-0582

## Abstract

Iron deficiency anaemia (IDA) currently affects 1.2 billion people and iron deficiency without anaemia (IDWA) is at least twice as common. IDWA is poorly recognised by clinicians despite its high prevalence, probably because of suboptimal screening recommendations. Diagnosing IDWA relies on a combination of tests, including haemoglobin and ferritin levels, as well as transferrin saturation. Although the causes of iron deficiency may sometimes be obvious, many tend to be overlooked. Iron sufficiency throughout pregnancy is necessary for maternal and foetal health. Preoperative IDWA must be corrected to reduce the risk of transfusion and postoperative anaemia. Oral iron is the first-line treatment for managing IDWA; however, intravenous supplementation should be used in chronic inflammatory conditions and when oral therapy is poorly tolerated or ineffective. This review considers the causes and clinical features of IDWA, calls for greater awareness of the condition, and proposes diagnostic and management algorithms.

**Keywords:** anaemia; iron deficiency; pregnancy; preoperative; transfusion.

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PubMed Disclaimer

# Iron Impacts on *Women*

There are studies that indicate the impact on low iron or suboptimal iron can extend past anema and present much earlier than the current cut offs.

The available evidence is mixed.

Studies available have small sample sizes and look at a limited period of time

There is insufficient information to state any of these findings as strong recommendations that should be applied to everyone

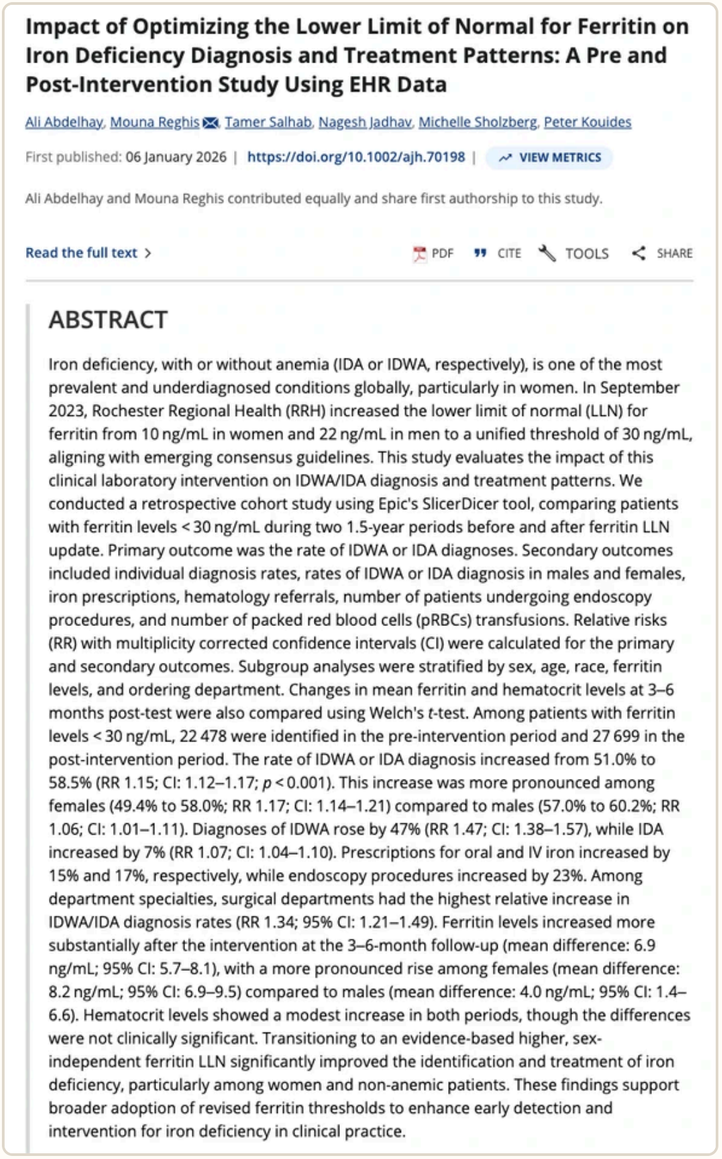
There is enough evidence to say that it is an area that should be explored further and should be considered as a part of a multimodal approach to a variety of concerns.

*(2026)*

# Impact of Optimizing the Lower Limit of Normal for Ferritin on Iron Deficiency Diagnosis and Treatment Patterns: A Pre & Post Intervention Study

globally under-diagnosed condition

adjusting the diagnostic parameters affected women more than men



**Impact of Optimizing the Lower Limit of Normal for Ferritin on Iron Deficiency Diagnosis and Treatment Patterns: A Pre and Post-Intervention Study Using EHR Data**

Ali Abdelhay, Mouna Reghis, Tamer Salhab, Nagesh Jadhav, Michelle Sholzberg, Peter Kouides

First published: 06 January 2026 | https://doi.org/10.1002/ajh.70198 | VIEW METRICS

Ali Abdelhay and Mouna Reghis contributed equally and share first authorship to this study.

Read the full text > PDF CITE TOOLS SHARE

ABSTRACT

Iron deficiency, with or without anemia (IDA or IDWA, respectively), is one of the most prevalent and underdiagnosed conditions globally, particularly in women. In September 2023, Rochester Regional Health (RRH) increased the lower limit of normal (LLN) for ferritin from 10 ng/mL in women and 22 ng/mL in men to a unified threshold of 30 ng/mL, aligning with emerging consensus guidelines. This study evaluates the impact of this clinical laboratory intervention on IDWA/IDA diagnosis and treatment patterns. We conducted a retrospective cohort study using Epic's SlicerDicer tool, comparing patients with ferritin levels < 30 ng/mL during two 1.5-year periods before and after ferritin LLN update. Primary outcome was the rate of IDWA or IDA diagnoses. Secondary outcomes included individual diagnosis rates, rates of IDWA or IDA diagnosis in males and females, iron prescriptions, hematology referrals, number of patients undergoing endoscopy procedures, and number of packed red blood cells (pRBCs) transfusions. Relative risks (RR) with multiplicity corrected confidence intervals (CI) were calculated for the primary and secondary outcomes. Subgroup analyses were stratified by sex, age, race, ferritin levels, and ordering department. Changes in mean ferritin and hematocrit levels at 3–6 months post-test were also compared using Welch's *t*-test. Among patients with ferritin levels < 30 ng/mL, 22 478 were identified in the pre-intervention period and 27 699 in the post-intervention period. The rate of IDWA or IDA diagnosis increased from 51.0% to 58.5% (RR 1.15; CI: 1.12–1.17; $p < 0.001$). This increase was more pronounced among females (49.4% to 58.0%; RR 1.17; CI: 1.14–1.21) compared to males (57.0% to 60.2%; RR 1.06; CI: 1.01–1.11). Diagnoses of IDWA rose by 47% (RR 1.47; CI: 1.38–1.57), while IDA increased by 7% (RR 1.07; CI: 1.04–1.10). Prescriptions for oral and IV iron increased by 15% and 17%, respectively, while endoscopy procedures increased by 23%. Among department specialties, surgical departments had the highest relative increase in IDWA/IDA diagnosis rates (RR 1.34; 95% CI: 1.21–1.49). Ferritin levels increased more substantially after the intervention at the 3–6-month follow-up (mean difference: 6.9 ng/mL; 95% CI: 5.7–8.1), with a more pronounced rise among females (mean difference: 8.2 ng/mL; 95% CI: 6.9–9.5) compared to males (mean difference: 4.0 ng/mL; 95% CI: 1.4–6.6). Hematocrit levels showed a modest increase in both periods, though the differences were not clinically significant. Transitioning to an evidence-based higher, sex-independent ferritin LLN significantly improved the identification and treatment of iron deficiency, particularly among women and non-anemic patients. These findings support broader adoption of revised ferritin thresholds to enhance early detection and intervention for iron deficiency in clinical practice.

# JObs of iron/ferritin

## *Iron*

- DNA Synthesis
- Mylination
- Energy Metabolism
- Neurotransmitter Synthesis
- Oxygen transport & storage
- Hematopoeisis
- Mitocondrial energy production
- Cellular proliferation - including osteoblast production

## *Ferritin*

- Iron storage
- Bioavailability buffer
- Cytoprotective detoxification
- Inflammatory marker - inflammation increases it’s production
- Protein
- Principle iron storage and detoxification protein
- Critically important in making iron both useable and non toxic

*(2026)*

# Cyclical Iron Homeostasis: The Role of Sex Hormones in Female Iron Regulation

Estrogen & Progesterone directly impact the degree of iron availability in the body

Iron follows a similar cyclical change

Full implications are unknown but it does tell us that states like perimenopause can substantailly change iron status



Review > Front Biosci (Landmark Ed). 2026 Sep 18;31(9):54404. doi: 10.31083/FBL54404.

**Cyclical Iron Homeostasis: The Role of Sex Hormones in Female Iron Regulation**

Darsiha Balakirishnan [1], Fanija Dukovski [1], Heather Bytheway [1], Cory Dugan [1 2 3]

Affiliations + expand

PMID: 42811967 DOI: 10.31083/FBL54404

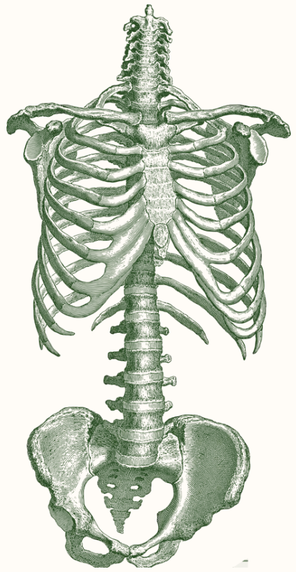
Free article

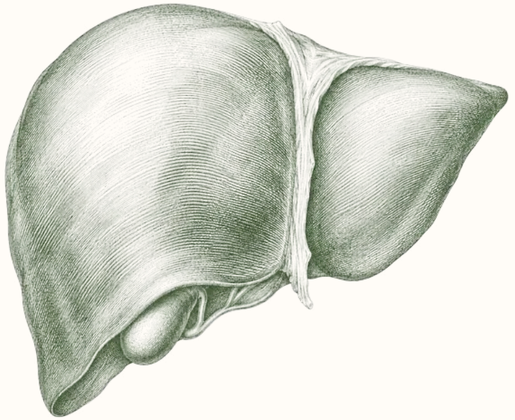
**Abstract**

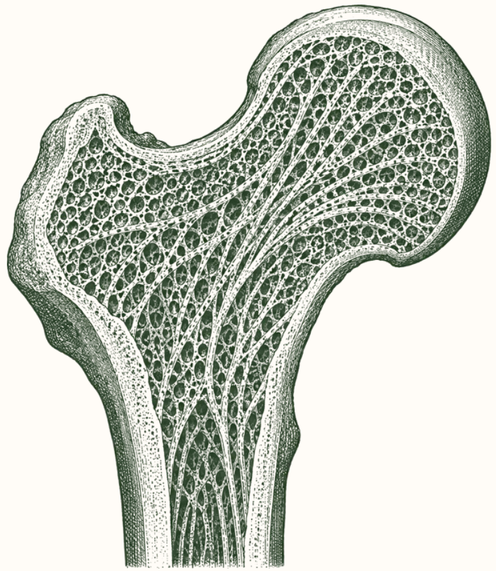
Iron homeostasis in premenopausal women is governed by a dual regulatory architecture that has been incompletely characterized in the clinical literature. The hepcidin-ferroportin axis, established as the master regulator of systemic iron flux, operates within a second-order hormonal framework in females of reproductive age: 17β-estradiol (E2) and progesterone (P4) directly modulate hepatic hepcidin transcription, superimposing a predictable, cycle-dependent oscillation on baseline iron availability. This review summarizes the molecular mechanisms underlying this cyclical regulation. E2 suppresses hepcidin expression through estrogen receptor-dependent regulation of the *HAMP* gene, mapped in hepatocyte models to an estrogen-responsive element (ERE) half-site. Preclinical evidence also implicates a competing pathway, mediated by the G protein-coupled estrogen receptor 1 (GPR30, also designated *GPER1*) and bone morphogenetic protein 6 (BMP6), with opposing upregulatory effects on hepcidin. Progesterone has an opposing effect, raising hepcidin in women when administered exogenously, an effect attributed in preclinical models to the progesterone receptor membrane component-1 (PGRMC1). When mapped onto the menstrual cycle, these hormonal actions predict a follicular phase of enhanced iron acquisition and a luteal phase of iron restriction. We distinguish that prediction from the evidence: the predicted late luteal hepcidin maximum is not evident in cohorts sampled serially across the cycle, and circulating interleukin-6 does not rise in the luteal phase, so the cycle-level consequence of the hormonal mechanism remains the central open question. The clinical implications extend to diagnostics and therapeutics: standard iron status biomarkers fluctuate predictably with cycle phase, confounding single-time-point assessments; and the clinical significance of these fluctuations differs mechanistically depending on whether the underlying deficit reflects absolute storage depletion, impaired functional delivery, or erythropoietic cellular restriction. This review proposes a mechanism-oriented biomarker subtyping framework to address this diagnostic gap and discusses cycle-synchronized iron supplementation as a biologically plausible hypothesis that warrants rigorous evaluation in future clinical trials.

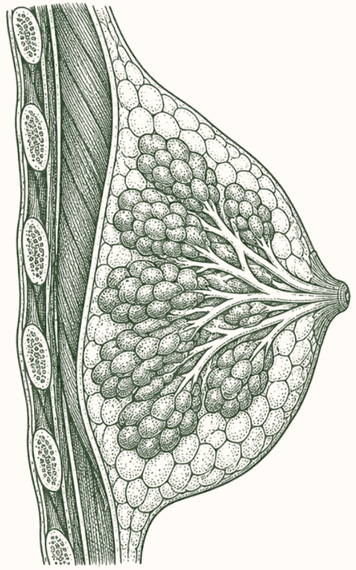
**Keywords:** estrogens; ferritins; hepcidins; iron; iron deficiencies; menstrual cycle; progesterone; transferrin; women's health.

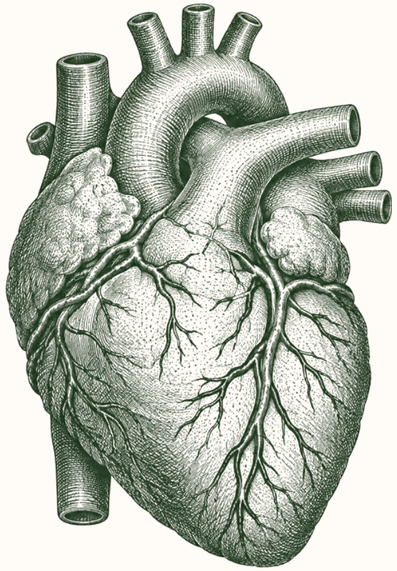
# High *Iron*

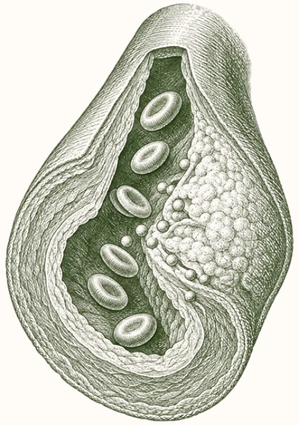




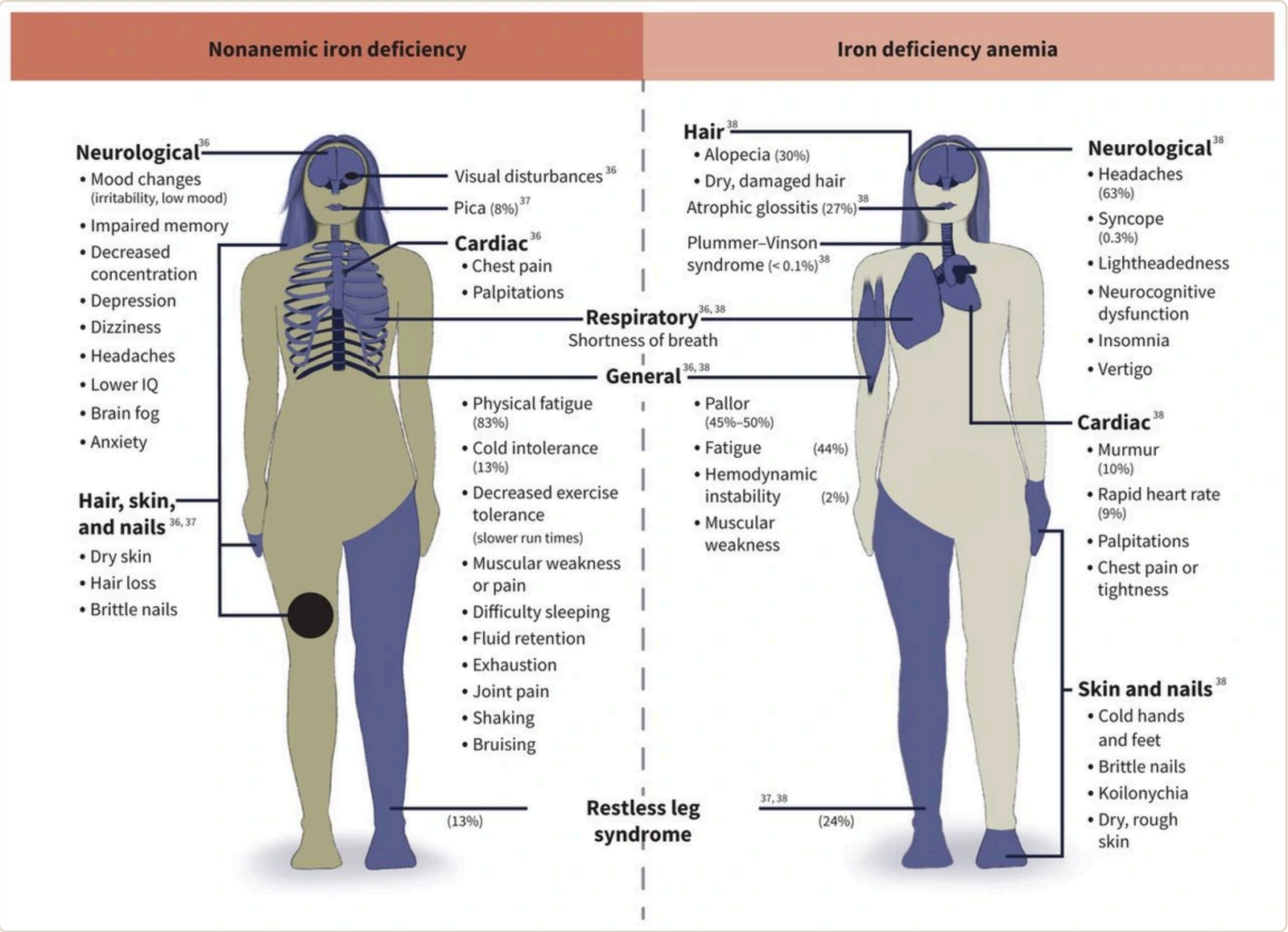








# Low *Iron*


Nonanemic iron deficiency
Iron deficiency anemia
Neurological[36]
• Mood changes (irritability, low mood)
• Impaired memory
• Decreased concentration
• Depression
• Dizziness
• Headaches
• Lower IQ
• Brain fog
• Anxiety
Hair, skin, and nails[36, 37]
• Dry skin
• Hair loss
• Brittle nails
Visual disturbances[36]
Pica (8%)[37]
Cardiac[36]
• Chest pain
• Palpitations
Respiratory[36, 38]
Shortness of breath
General[36, 38]
• Physical fatigue (83%)
• Cold intolerance (13%)
• Decreased exercise tolerance (slower run times)
• Muscular weakness or pain
• Difficulty sleeping
• Fluid retention
• Exhaustion
• Joint pain
• Shaking
• Bruising
(13%)
Restless leg syndrome
Hair[38]
• Alopecia (30%)
• Dry, damaged hair
Atrophic glossitis (27%)[38]
Plummer–Vinson syndrome (< 0.1%)[38]
• Pallor (45%–50%)
• Fatigue (44%)
• Hemodynamic instability (2%)
• Muscular weakness
37, 38
(24%)
Neurological[38]
• Headaches (63%)
• Syncope (0.3%)
• Lightheadedness
• Neurocognitive dysfunction
• Insomnia
• Vertigo
Cardiac[38]
• Murmur (10%)
• Rapid heart rate (9%)
• Palpitations
• Chest pain or tightness
Skin and nails[38]
• Cold hands and feet
• Brittle nails
• Koilonychia
• Dry, rough skin

*(2024)*

# The Complementary Roles of Iron and Estrogen in Menopausal Differences in Cardiometabolic Outcomes

iron metabolism changes with hormonal changes in menopause

after menopause, excessive iron can increase cardiometabolic risks

Iron stores are lower than men’s before menopause but higher after



Review > Clin Nutr. 2024 May;43(5):1136-1150. doi: 10.1016/j.clnu.2024.03.026.
Epub 2024 Apr 3.

**The complementary roles of iron and estrogen in menopausal differences in cardiometabolic outcomes**

Noushin Sadat Ahanchi [1], Farnaz Khatami [2], Erand Llanaj [3], Hugo G Quezada-Pinedo [4], Helga Dizdari [5], Arjola Bano [6], Marija Glisic [7], Michele F Eisenga [8], Pedro-Marques Vidal [9], Taulant Muka [10]

Affiliations + expand

PMID: 38593499 DOI: 10.1016/j.clnu.2024.03.026

Free article

**Abstract**

Biological hormonal changes are frequently cited as an explanatory factor of sex and menopause differences in cardiometabolic diseases (CMD) and its associated risk factors. However, iron metabolism which varies between sexes and among women of different reproductive stages could also play a role. Recent evidence suggest that iron may contribute to CMD risk by modulating oxidative stress pathways and inflammatory responses, offering insights into the mechanistic interplay between iron and CMD development. In the current review, we provide a critical appraisal of the existing evidence on sex and menopausal differences in CMD, discuss the pitfall of current estrogen hypothesis as sole explanation, and the emerging role of iron in CMD as complementary pathway. Prior to menopause, body iron stores are lower in females as compared to males, but the increase during and after menopause, is tandem with an increased CMD risk. Importantly, basic science experiments show that an increased iron status is related to the development of type 2 diabetes (T2D), and different cardiovascular diseases (CVD). While epidemiological studies have consistently reported associations between heme iron intake and some iron biomarkers such as ferritin and transferrin saturation with the risk of T2D, the evidence regarding their connection to CVD remains controversial. We delve into the factors contributing to this inconsistency, and the limitation of relying on observational evidence, as it does not necessarily imply causation. In conclusion, we provide recommendations for future studies on evaluating the potential role of iron in elucidating the sex and menopausal differences observed in CMD.

**Keywords:** Cardiometabolic risk factors; Iron biomarkers; Menopause; Sex.

*(2025)*

# Psychiatric and cognitive outcomes of iron supplementation in non-anemic children, adolescents, and menstruating adults: a meta analysis and systemic review

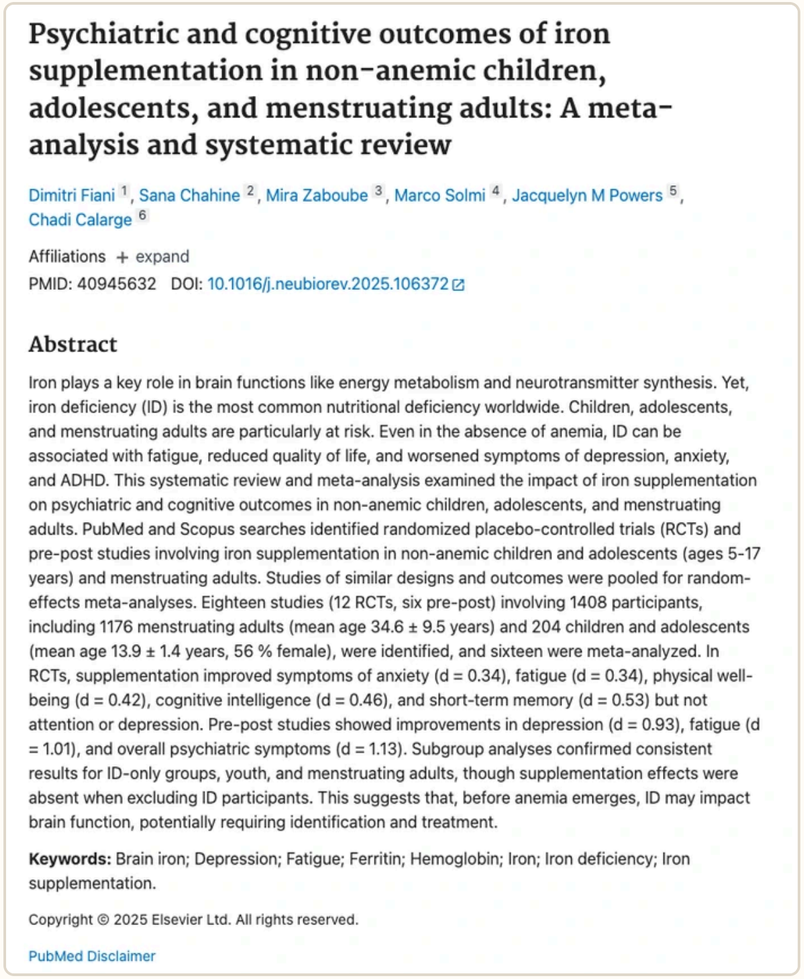
Despite iron being critical in energy metabolism and neurotransmitter synthesis, Iron deficiency is the most common nutritional deficiency world wide

Meta analysis looked at 18 studies looking at iron status on psychiatric and cognitive outcomes

Supplementation improved: anxiety, fatigue, wellbeing, cognitive intelligence and short term memory

No improvement in attention.

Benefits of supplementation were not found in adult males



**Psychiatric and cognitive outcomes of iron supplementation in non-anemic children, adolescents, and menstruating adults: A meta-analysis and systematic review**

Dimitri Fiani [1], Sana Chahine [2], Mira Zaboube [3], Marco Solmi [4], Jacquelyn M Powers [5], Chadi Calarge [6]

Affiliations + expand

PMID: 40945632 DOI: 10.1016/j.neubiorev.2025.106372

**Abstract**

Iron plays a key role in brain functions like energy metabolism and neurotransmitter synthesis. Yet, iron deficiency (ID) is the most common nutritional deficiency worldwide. Children, adolescents, and menstruating adults are particularly at risk. Even in the absence of anemia, ID can be associated with fatigue, reduced quality of life, and worsened symptoms of depression, anxiety, and ADHD. This systematic review and meta-analysis examined the impact of iron supplementation on psychiatric and cognitive outcomes in non-anemic children, adolescents, and menstruating adults. PubMed and Scopus searches identified randomized placebo-controlled trials (RCTs) and pre-post studies involving iron supplementation in non-anemic children and adolescents (ages 5-17 years) and menstruating adults. Studies of similar designs and outcomes were pooled for random-effects meta-analyses. Eighteen studies (12 RCTs, six pre-post) involving 1408 participants, including 1176 menstruating adults (mean age 34.6 ± 9.5 years) and 204 children and adolescents (mean age 13.9 ± 1.4 years, 56 % female), were identified, and sixteen were meta-analyzed. In RCTs, supplementation improved symptoms of anxiety (d = 0.34), fatigue (d = 0.34), physical well-being (d = 0.42), cognitive intelligence (d = 0.46), and short-term memory (d = 0.53) but not attention or depression. Pre-post studies showed improvements in depression (d = 0.93), fatigue (d = 1.01), and overall psychiatric symptoms (d = 1.13). Subgroup analyses confirmed consistent results for ID-only groups, youth, and menstruating adults, though supplementation effects were absent when excluding ID participants. This suggests that, before anemia emerges, ID may impact brain function, potentially requiring identification and treatment.

**Keywords:** Brain iron; Depression; Fatigue; Ferritin; Hemoglobin; Iron; Iron deficiency; Iron supplementation.

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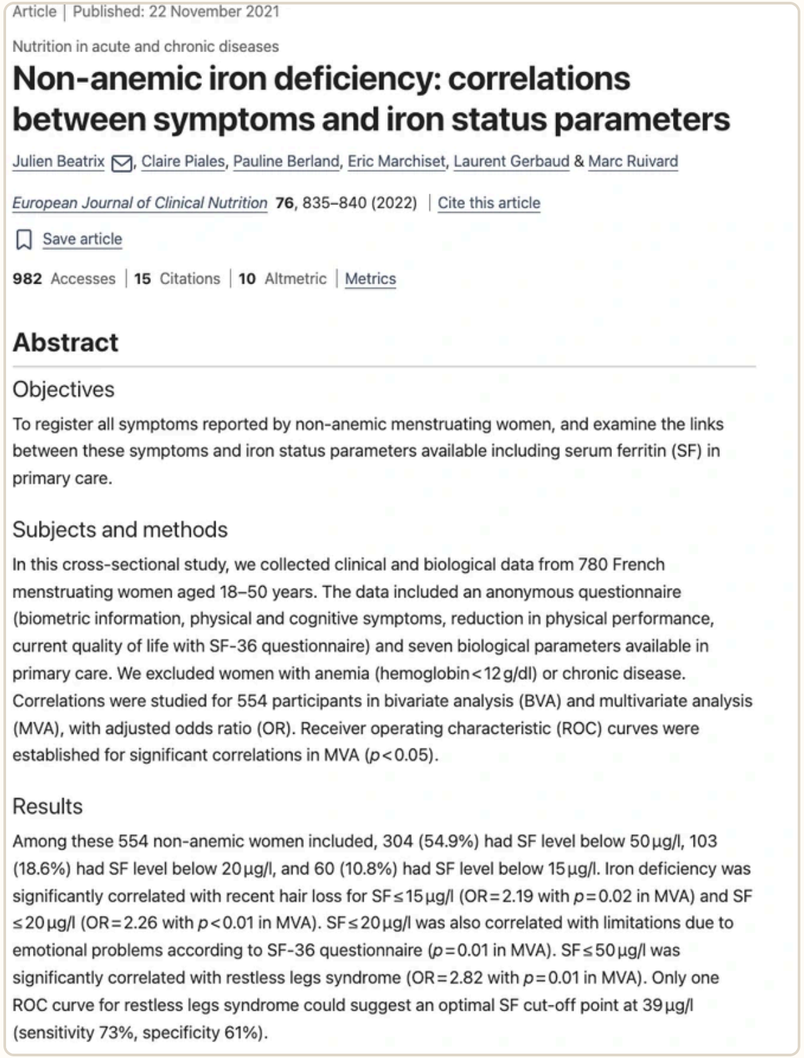
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*(2021)*

# Non-anemic iron deficiency: Correlations between symptoms and iron status parameters

554 non anemic women with varying Ferritin levels (<50, <20, <15)

Iron deficiency associated with symptoms after falling below 50



Article | Published: 22 November 2021

Nutrition in acute and chronic diseases

## Non-anemic iron deficiency: correlations between symptoms and iron status parameters

Julien Beatrix, Claire Piales, Pauline Berland, Eric Marchiset, Laurent Gerbaud & Marc Ruivard

*European Journal of Clinical Nutrition* **76**, 835–840 (2022) | Cite this article

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**982** Accesses | **15** Citations | **10** Altmetric | Metrics

### Abstract

Objectives

To register all symptoms reported by non-anemic menstruating women, and examine the links between these symptoms and iron status parameters available including serum ferritin (SF) in primary care.

Subjects and methods

In this cross-sectional study, we collected clinical and biological data from 780 French menstruating women aged 18–50 years. The data included an anonymous questionnaire (biometric information, physical and cognitive symptoms, reduction in physical performance, current quality of life with SF-36 questionnaire) and seven biological parameters available in primary care. We excluded women with anemia (hemoglobin < 12 g/dl) or chronic disease. Correlations were studied for 554 participants in bivariate analysis (BVA) and multivariate analysis (MVA), with adjusted odds ratio (OR). Receiver operating characteristic (ROC) curves were established for significant correlations in MVA ($p < 0.05$).

Results

Among these 554 non-anemic women included, 304 (54.9%) had SF level below 50 µg/l, 103 (18.6%) had SF level below 20 µg/l, and 60 (10.8%) had SF level below 15 µg/l. Iron deficiency was significantly correlated with recent hair loss for SF ≤ 15 µg/l (OR = 2.19 with $p = 0.02$ in MVA) and SF ≤ 20 µg/l (OR = 2.26 with $p < 0.01$ in MVA). SF ≤ 20 µg/l was also correlated with limitations due to emotional problems according to SF-36 questionnaire ($p = 0.01$ in MVA). SF ≤ 50 µg/l was significantly correlated with restless legs syndrome (OR = 2.82 with $p = 0.01$ in MVA). Only one ROC curve for restless legs syndrome could suggest an optimal SF cut-off point at 39 µg/l (sensitivity 73%, specificity 61%).

*(2019)*

# Mild iron deficiency may affect female endurance and behavior.

- Central Nervous System requires iron for functioning
- Iron status was compared to perceived energy level and EEG findings
- 23 women between 20-32 in their follicular phase
- Those with lower ferritin levels reported lower activity and endurance
- The groups did not differ in intelligence however, the deficient group required more time to complete the assessment



› Physiol Behav. 2019 Jun 1:205:44-50. doi: 10.1016/j.physbeh.2018.09.012. Epub 2018 Sep 26.

**Mild iron deficiency may affect female endurance and behavior**

Inga Dziembowska [1], Justyna Kwapisz [2], Paweł Izdebski [3], Ewa Żekanowska [2]

Affiliations + expand

PMID: 30267737 DOI: 10.1016/j.physbeh.2018.09.012

**Abstract**

The function of central nervous system (CNS) processes is highly dependent on iron containing enzymes and proteins. Therefore, even mild iron deficiency (ID) may result in decreased endurance, increased irritability and withdrawal behavior among women. The current study was designed to assess the effects of iron status on perceived energy level along with its electroencephalographical (EEG) correlates and cognitive capacity of young women, since this group is at particular risk of experiencing depleted iron stores and iron deficiency anemia. The study group consisted of 23 non-anemic women of reproductive age (20-32 years) during their follicular phase of menstrual cycle. All participants were moderately physically active with a mean weekly energy expenditure of 1110 ± 450 Metabolic Equivalent of Task per minute-(MET/min). The group's electroencephalographic (EEG) alpha asymmetry and psychometric data - intelligence, anxiety level and temperamental traits (activity, briskness, sensory sensitivity, emotional reactivity, perseveration and endurance) were assessed and compared between iron deficient (serum ferritin <12 μg/l) and iron sufficient (serum ferritin ≥12 μg/l) females. Participants with depleted iron stores reported lower levels of activity ($p = .004$; partial $\eta^2 = 0.34$) and endurance ($p = .038$, partial $\eta^2 = 0.25$), temperamental traits, and presented increased relative left EEG alpha activity in prefrontal regions ($p = .004$; partial $\eta^2 = 0.63$), which is a characteristic EEG pattern for withdrawal tendencies. Iron-depleted females did not differ in anxiety and general intelligence scores from those which were iron sufficient, however needed more time (median difference: 3 min, $p = .004$; partial $\eta^2 = 0.53$) to complete an administered intelligence test, which may reflect lowered endurance during a cognitive-demanding task. These data provide proof that that even mild iron deficiency may be connected with CNS function and lead to decreased endurance, or at least cognitive endurance, which may be a risk-factor for mood disorders. Therefore, more attention should be paid to preventing even mild iron deficiency, particularly among young women.

**Keywords:** Anxiety; Cognition; EEG alpha asymmetry; Endurance; Iron deficiency.

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*(2016)*

# Iron Status Is associated with performance on executive functioning tasks in nonanemic young women.

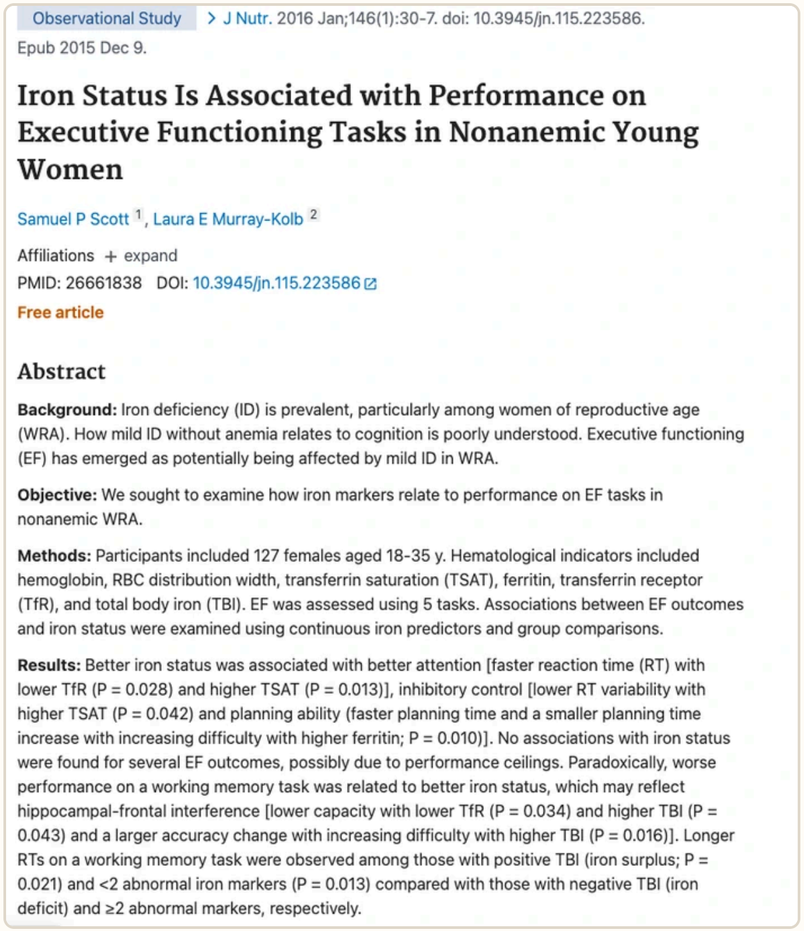
127 females between 18-54 years

Iron status and executive funcitoning was measured

Better iron status was associated with better attention (faster reaction time), inhibitory control, planning ability, and requiring less planning time,

Other executive functions had no change, and working memory appeared to worsen with better iron status.

Limited study



Observational Study > J Nutr. 2016 Jan;146(1):30-7. doi: 10.3945/jn.115.223586.
Epub 2015 Dec 9.

**Iron Status Is Associated with Performance on Executive Functioning Tasks in Nonanemic Young Women**

Samuel P Scott [1], Laura E Murray-Kolb [2]

Affiliations + expand

PMID: 26661838 DOI: 10.3945/jn.115.223586

Free article

**Abstract**

**Background:** Iron deficiency (ID) is prevalent, particularly among women of reproductive age (WRA). How mild ID without anemia relates to cognition is poorly understood. Executive functioning (EF) has emerged as potentially being affected by mild ID in WRA.

**Objective:** We sought to examine how iron markers relate to performance on EF tasks in nonanemic WRA.

**Methods:** Participants included 127 females aged 18-35 y. Hematological indicators included hemoglobin, RBC distribution width, transferrin saturation (TSAT), ferritin, transferrin receptor (TfR), and total body iron (TBI). EF was assessed using 5 tasks. Associations between EF outcomes and iron status were examined using continuous iron predictors and group comparisons.

**Results:** Better iron status was associated with better attention [faster reaction time (RT) with lower TfR (P = 0.028) and higher TSAT (P = 0.013)], inhibitory control [lower RT variability with higher TSAT (P = 0.042) and planning ability (faster planning time and a smaller planning time increase with increasing difficulty with higher ferritin; P = 0.010)]. No associations with iron status were found for several EF outcomes, possibly due to performance ceilings. Paradoxically, worse performance on a working memory task was related to better iron status, which may reflect hippocampal-frontal interference [lower capacity with lower TfR (P = 0.034) and higher TBI (P = 0.043) and a larger accuracy change with increasing difficulty with higher TBI (P = 0.016)]. Longer RTs on a working memory task were observed among those with positive TBI (iron surplus; P = 0.021) and <2 abnormal iron markers (P = 0.013) compared with those with negative TBI (iron deficit) and ≥2 abnormal markers, respectively.

*(2011)*

# Iron status and neuropsychological consequences in women of reproductive age: What do we know and where are we headed?

iron status disproportionately affects infants, children, and women of reproductive age

indication to explore the interaction between iron status, cognition, emotions, and behavior

The question remains: What degree of iron deficiency associated with negative consequences?



Review > J Nutr. 2011 Apr 1;141(4):747S-755S. doi: 10.3945/jn.110.130658. Epub 2011 Feb 23.

**Iron status and neuropsychological consequences in women of reproductive age: what do we know and where are we headed?**

Laura E Murray-Kolb [1]

Affiliations + expand

PMID: 21346109 DOI: 10.3945/jn.110.130658

Free article

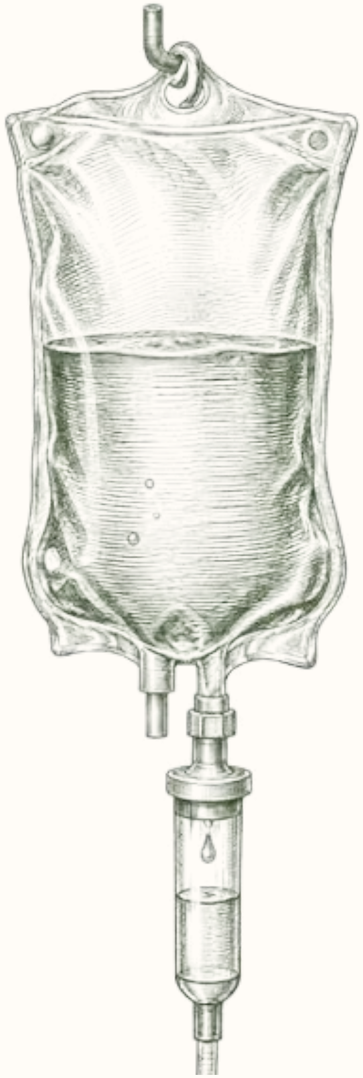
**Abstract**

Iron deficiency disproportionately affects infants, children, and women of reproductive age. It is more prevalent in developing countries but continues to be a problem in developed countries. Included among the consequences of iron deficiency are changes in cognitive performance, emotions, and behavior. Although the behavioral and developmental sequelae of iron deficiency in young children have received much interest, data on the consequences of iron deficiency in women of reproductive age are just beginning to emerge. Interest in this area increased as a result of the findings that brain iron is much more fluid than previously thought and iron-deficient animals experience nondevelopment-dependent brain alterations. These findings, coupled with the symptoms often reported by iron-deficient women (lethargy, inability to concentrate, difficulty with memory, etc.), have led researchers to begin to explore the relation between iron and neuropsychological outcomes in this population. This paper reviews findings from studies that have examined the brain functional consequences of iron deficiency in women of reproductive age. Additional consideration is given to the fact that detriments to cognition and behavior in a woman of reproductive age may have negative implications beyond the woman's health; specifically, they may negatively influence her children. Therefore, the roles that cognitive and behavioral changes in these women have on their children's development is also reviewed. The paper concludes with a discussion on ways to use technological advances (especially in neuroimaging techniques) to help us elucidate the role of iron in the brains of women of reproductive age.

PubMed Disclaimer



VS



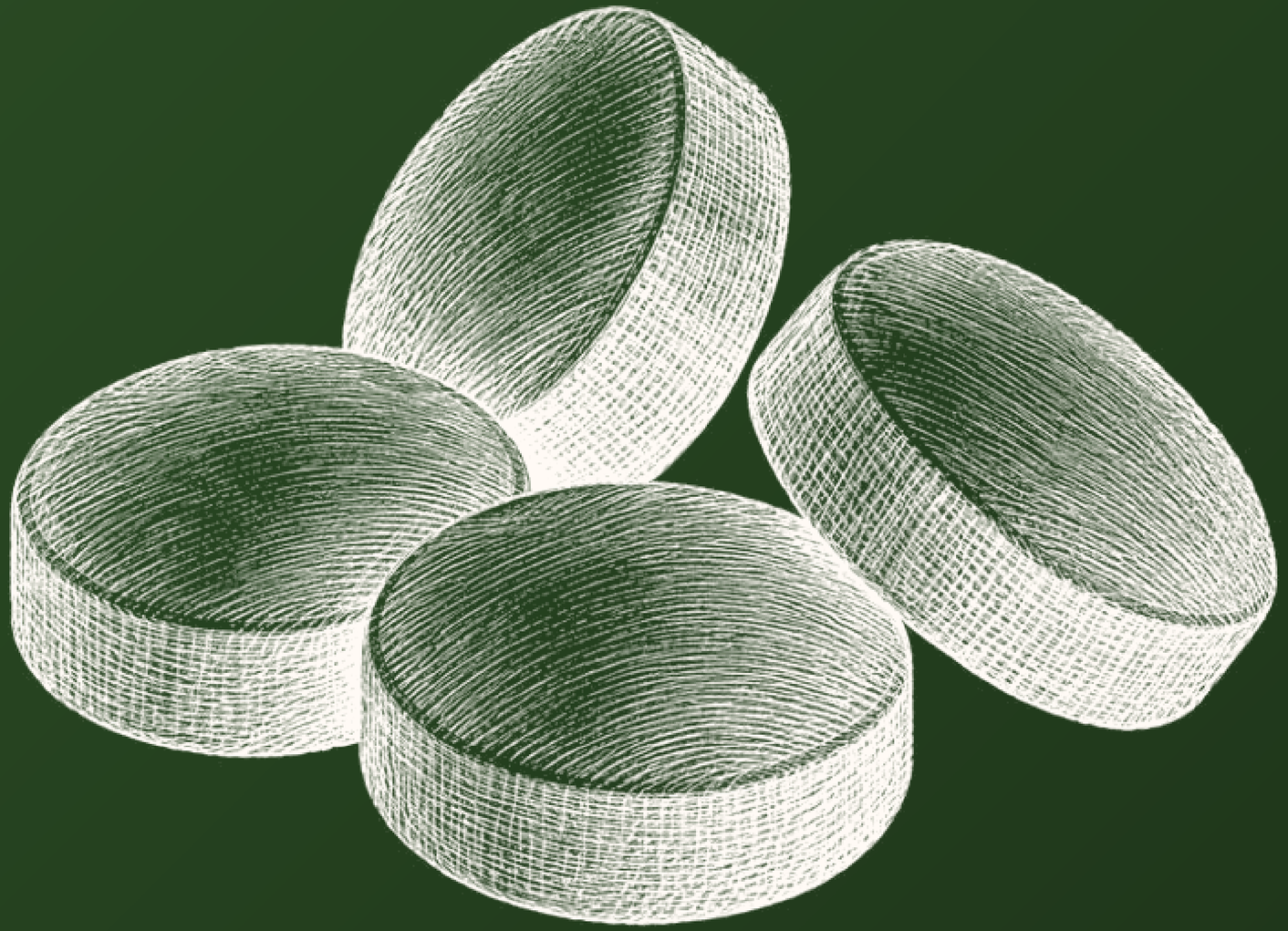
# dietary *iron*

maintaning iron stores requires at least 20mg elemental iron daily

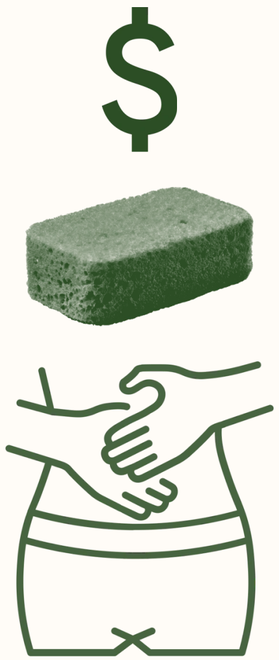
insufficient to correct deficiency

animal sources have heme iron and are better absorbed

Heme iron is not affected by other foods eaten together, non heme iron absorption is greatly affected by cocurrent foods



# cokmpare ferr *salts*

| Sulfate | Fumurate | Gluconate | Bilycinate | Polysaccaride Complex | Maltol |
| --- | --- | --- | --- | --- | --- |
| |  | |  |  |  |

# *Recommendations*

Determine elemental iron dose

aim for 60-110 elemental iron daily

Empty stomach. Avoid with calcium/magnesium

With Vitamin C

Common Side Effects

